

Data Handling and Infrastructure for AI

Module induction

- Advanced module in AI/ML data handling and infrastructure
- Focused on turning experiments into reliable AI systems
- Blends concepts, platforms, and practical engineering decisions

Why this module matters

Modern AI systems need more than a model:

- high-quality data
- repeatable pipelines
- suitable storage
- monitoring and versioning
- scalable compute and cloud infrastructure

This module is about the foundations that make AI systems usable in practice.

What the module covers

- Data types: tabular, graph, and unstructured
- Feature engineering, statistics, and analysis
- Notebooks, ETL, and batch vs streaming pipelines
- Data storage: SQL, NoSQL, lakes, and warehouses
- AI/ML Ops: versioning, monitoring, and CI/CD
- HPC for AI: distributed systems, GPUs, and TPUs
- Cloud platforms for AI delivery
- Infrastructure patterns for scalable LLM training

From experimentation to production

The module follows a practical progression:

1. Understand the data
2. Design transformations and workflows
3. Choose storage and compute appropriately
4. Operationalise with AI/ML Ops practices
5. Scale on cloud and high-performance infrastructure

Goal: build systems that are robust, efficient, and maintainable.

Learning outcomes

By the end of the module, you should be able to:

1. Compare different data types for AI/ML use cases
2. Explain the role of feature engineering and statistics
3. Design data pipelines and distinguish batch from streaming
4. Evaluate modern GenAI patterns in applied projects

Learning outcomes continued

You should also be able to:

- 5. Compare storage options for different AI/ML workloads
- 6. Apply versioning, monitoring, and CI/CD practices
- 7. Use HPC resources for data processing and training
- 8. Propose scalable cloud infrastructure strategies for LLMs

How you will learn

- Live online lectures for core ideas and framing
- Self-paced practical labs with lecturer support
- Case studies drawn from deep learning and GenAI practice
- Online channels for lecturer and peer support
- Asynchronous materials for independent study

Expect both conceptual understanding and applied technical work.

Assessment snapshot

Continuous assessment = **100%**

- **In-class assessment (50%)**
 - aligned with outcomes 1, 2, and 3
- **Assignment (50%)**
 - aligned with outcomes 1, 4, and 5

Success depends on sound design choices, not just working code.

Tools and platforms

Across the module you may work with or discuss:

- Jupyter Notebook environments
- ETL and workflow patterns
- SQL and NoSQL systems
- Data lakes and cloud warehouses
- Hugging Face and Kaggle style ecosystems
- PyTorch-based large-scale workflows
- AWS, GCP, and Azure
- GPU and TPU-backed compute

How to succeed in this module

Build strong habits from the start:

- **Data awareness** - know the structure and quality of your data
- **Pipeline thinking** - make work reproducible and traceable
- **Operational discipline** - version, monitor, and document decisions
- **Infrastructure judgement** - balance scale, performance, and cost

Think like an engineer as well as a model builder.

Takeaway

This module is about building the systems around AI:

- data
- workflows
- storage
- operations
- compute
- cloud infrastructure

If you can make AI solutions repeatable, scalable, and maintainable, you are doing the real work of delivery.